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LESSON 1: Library management system

--Identify candidate classes (Books, Readers, Vouchers, Employees, Publishers)

+ Book: main entity, managing book information.

+ DocGia: person who borrows books.

+ PhieuMuon: records borrowing/returning.

+ NhanVien: system manager (librarian, warehouse keeper...).

+ NhaXuatBan: book source information.

- Properties and behavior (methods) of each class

+ Sach

Attributes: maSach (PK, string), nameSach (string), tacGia (string),
namXuatBan (int), manNXB (FK), soLuongTon (int), trangThai (string: "Yes
Behavior: addSach(), capNhatSach(), deleteSach(), saveSach(),

+ DocGia

Attributes: maDocGia (PK, string), hoTen (string), birth date (date), diaChi
Behavior: dangKyThe(), capNhatThongTin(), viewLichSuMuon().

Attributes: maPhieu (PK, string), maDocGia (FK), maSach (FK),
maNhanVien (FK), nayMuon (date), nayHenTra (date), nayTraThucTe
Behavior: createPhieuMuon(), searchSach(), saveTraQuaHan(), crystalPhieuPhat().

Attributes: maNhanVien (PK, string), hoTen (string), chucVu (string),
Behavior: dangNhap(), quanLyDocGia(), quanLyMuonTra(), quanLySach

+ NhaXuatBan

Attributes: maNXB (PK, string), nameNXB (string), diaChi (string), email
Behavior: addNXB(), capNhatNXB().

- Relationships

□□ Books and Publishers: 1 book – 1

Publishing House

- Readers and Borrowing Slips: 1 reader – many borrowing slips
- Books and borrowing slips: 1 book – many borrowing slips
- Employees and Vouchers: 1 employee – many vouchers

LESSON 2: Online sales management system

- Identify main use cases

Main actors:

- + Customer (Customers / Potential Customers).
- + NguoiQuanLy / Admin (Administrator / Sales staff).

Main use cases (customer registration, ordering, payment, product management):

KhachHang:

- DangKyTaiKhan (Customer registration).
- DangNhap (Sign in).
- SeeSanPham (Search / View product list).
- ThemVaoGioHang (Add to cart).
- DatHang (Order).
- ThanhToan (Online payment / COD).
- View DonHang (View order history).

Admin / NguoiQuanLy:

- QuanLySanPham (Add/Edit/Delete products).
- QuanLyDonHang (View/Confirm/Cancel order).
- QuanLyKhachHang (Customer information management).
- QuanLyThanhToan (Payment processing, refund if necessary).

Extend / Include:

- ThanhToan includes XacNhanDonHang.
- DatHang extends ThanhToan (if paying online).

- From use case, find the necessary classes

- Customer (Customer): from register, view orders.
- SanPham (Product): product management.
- GioHang (Cart): temporarily store products before ordering.
- Don Hang (Order): from order, payment.
- ChiTietDonHang (OrderDetail): product details in the order.
- ThanhToan (Payment): payment processing (may have subclasses such as OnlinePayment, COD).
- Admin or Personnel (inherited from User if there is a user system).

- Analyze relationships between classes (specify inheritance if any)

□□KhachHang and DonHang: 1 customer – many orders

□□DonHang and ChiTietDonHang: 1 order – many order details (Composition)

□□ChiTietDonHang and SanPham: many single details – 1 product

□□DonHang and ThanhToan: 1 order – 1 payment

Admins are all the same type of User, sharing the same account, password, and full name).

Lesson 3: Convert class diagrams into data tables

1.MonHoc Table

No	School name	Data Types	Bind			
			PK	FK	NULL	Unique
1	maMon	Nvarchar(10)	x		x	x
2	tenMon	Nvarchar(50)			x	
3	soTinChiLT	Int			x	
4	soTinChiTH	Int			x	

2. Science Table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maKhoaHoc	Nvarchar(10)	x			x
2	tenKhoaHoc	Nvarchar(50)			x	
3	ngayTao	Date			x	